AI Usage DISCLOSURE FORM

1. **Team Details**

Team Name: Red-Claw

Project / Product Name: Reddit Business Query Researcher

Organization / Institution (if any): RV College of Engineering

Submission Date: 8/5/2026

2. **AI Usage Declaration**

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

Yes

3. **Purpose of AI Usage (Brief Details)**

Idea generation / brainstorming - None

Code generation or assistance – Used Claude to set up discord webhook to read from forms and used it to link the discord form reader to reddit read-only skill.

UI / UX design - None

Content creation – Used Gemini to create in-detailed workflow and presentation to capture the project deliverable effectively.

Data analysis - None

Testing / debugging – OpenClaw debugged heartbeat issues and fixed its own code along with manual checking.

Other – Copilot for code completion and verification.

4. **Feature Origin Classification**

a) **Feature Name –** Discord Form Monitoring & Message Ingestion

Self-Generated/AI-Generated/Both – Both

Description (include AI tools/platform Used, prompt used, Output Summary, Modification) –

Built using Discord API integration and OpenClaw runtime. AI assistance was used for workflow structuring and event-driven architecture suggestions using ChatGPT/Groq. Prompt used: *"Design a Discord-based agent system that listens to form submissions and triggers downstream AI workflows."* Output included webhook/event polling architecture and session routing ideas. Modified by implementing custom heartbeat polling every 20 seconds, session handling, and Discord-specific message validation.

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b) **Feature Name** – OpenClaw 5-Layer Agentic Architecture

Self-Generated/AI-Generated/Both – Both

Description –
Architecture was conceptualized manually and refined using AI tools like ChatGPT and Claude for control-plane layering ideas. Prompt used: *"Create a scalable 5-layer AI agent architecture for autonomous workflow execution."* AI generated modular layer suggestions including gateway, execution engine, and skill separation. Final architecture was customized into Communication Layer, Adapter Layer, Gateway Layer, Pi Engine, and Skill Execution Layer.

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c) **Feature Name** – Discord Form Reader Skill

Self-Generated/AI-Generated/Both – Both

Description –
Developed using Node.js and OpenClaw skill framework. AI assistance from ChatGPT was used to generate parsing logic and JSON schema suggestions. Prompt used: *"Generate a parser that extracts structured business data from Discord form submissions."* Output included regex-based extraction and schema validation logic. Modified to support specific fields like business name, industry, audience, products/services, and USP.

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d) **Feature Name** – Reddit Research Pipeline Orchestration

Self-Generated/AI-Generated/Both – Both

Description –
Built using custom orchestration scripts and OpenClaw skills. AI tools like Gemini and ChatGPT were used to improve pipeline sequencing and workflow chaining. Prompt used: *"Design a workflow that converts structured business forms into Reddit market research insights."* Output generated orchestration flow ideas connecting form parsing, subreddit search, trend extraction, and reporting. Modified to integrate seamlessly with Discord response workflows.

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e) **Feature Name –** Reddit Research & Trend Extraction

Self-Generated/AI-Generated/Both – Both

Description –
Implemented using Reddit public API endpoints with custom Node.js scripts. AI assistance was used for subreddit ranking logic and insight extraction prompts. Prompt used: *"Find relevant Reddit communities and summarize trending pain points for SaaS businesses."* AI generated summaries, subreddit recommendations, and discussion themes. Final system was modified to support engagement filtering, trend detection, and structured reporting.

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f) **Feature Name –** AI-Powered Insight Summarization

Self-Generated/AI-Generated/Both – AI-Generated

Description –
Used DeepSeek R1 via Groq and Gemini models for summarizing Reddit discussions into concise business insights. Prompt used: *"Summarize Reddit discussions into actionable marketing insights and recurring customer pain points."* Output included trend summaries, recommendations, and opportunity analysis. Modified by adding formatting rules and subreddit citation requirements.

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g) **Feature Name –** Pi Engine Reasoning Layer

Self-Generated/AI-Generated/Both – Both

Description –
Built using OpenClaw runtime with personality/context control through SOUL.md and AGENTS.md. AI assistance was used to structure reasoning flows and memory management. Prompt used: *"Create an AI agent personality and rule system for a Reddit research assistant."* Output included behavioral guidelines, execution rules, and identity structure. Modified heavily to fit the Lob 🦞 research assistant persona.

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h) **Feature Name –** Session & Memory Management

Self-Generated/AI-Generated/Both – Self-Generated

Description –
Implemented using JSONL session storage and OpenClaw session architecture. No direct AI generation was used for core persistence logic. Session tracking, activity logging, and workflow state handling were custom-developed and optimized for long-running research workflows.

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i) **Feature Name** – Automated Heartbeat Polling System

Self-Generated/AI-Generated/Both – Both

Description –
Built using OpenClaw heartbeat configuration and custom polling logic. AI tools were used for scheduling and automation recommendations. Prompt used: *"Design a lightweight polling mechanism for periodically checking Discord channels."* Output included periodic workflow execution concepts. Modified to run every 20 seconds with optimized workflow triggering.

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j) **Feature Name** – Gateway & API Routing Layer

Self-Generated/AI-Generated/Both – Both

Description –
Implemented using OpenClaw gateway architecture on port 18789. AI assistance was used for designing API structures and routing logic. Prompt used: *"Generate REST API endpoints for AI agent communication and session management."* Output included health check APIs, message submission endpoints, and session retrieval endpoints. Modified to support Discord integration and internal agent routing.

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k) **Feature Name** – Structured Research Report Generation

Self-Generated/AI-Generated/Both – Both

Description –
Generated using LLM summarization combined with manually designed JSON schemas and formatting rules. Prompt used: *"Create a concise research report summarizing subreddit trends, high-engagement posts, and recommendations."* Output included formatted business intelligence reports. Modified to improve readability and Discord compatibility.

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l) **Feature Name** – Multi-Model AI Integration

Self-Generated/AI-Generated/Both – Both

Description –
Integrated DeepSeek R1, Gemini, and model routing into OpenClaw runtime. AI assistance was used to compare models and optimize reasoning workflows. Prompt used: *"Compare LLM models suitable for Reddit research summarization and reasoning tasks."* Output helped determine best models for summarization, orchestration, and instruction-following. Final implementation added fallback routing and configurable model settings.

5. **Ethical & Compliance Confirmation**

AI usage complies with guidelines and policies. **Yes**

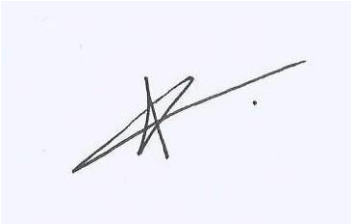
No proprietary or copyrighted data misused. **I Agree**

6. **Declaration & Sign-Off**

Name of Team Representative: Akshay Avinash

Role: Team Lead

Signature:



Date: 8/5/2026